
Introduction to Social Network Analysis

Concepts, Data, and Visualization

New College of Florida • Fall 2026

Instructor

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Office

Announced on Canvas

Office Hours

Tuesday and Thursday, 10:00–11:00 a.m.;

Tuesday, 2:00–3:00 p.m.; and by appointment

Meeting Time

Monday and Thursday, 4:00–5:20 p.m.

Location

See the official course schedule and Canvas

Canvas

Course announcements, readings, lab modules, data, and R scripts will be posted on the course site

Course level: Upper-level undergraduate. **Audience:** Political science, public policy, sociology, psychology, economics, history, anthropology, international and area studies, and related AOCs.

Course Overview

In fifteenth-century Florence, the Medici rose to power without holding the most wealth or the highest offices. Their advantage was structural: they occupied a pivotal position in a web of marriage and business ties. In a contemporary high school, networks recorded through sensors, contact diaries, and friendship surveys reveal strikingly different pictures of the same social world. In the U.S. Congress, patterns of agreement across roll-call votes disclose coalitions and divisions that no organizational chart can show.

None of these findings comes from studying individuals one at a time. They require looking at **relations**—who is connected to whom, and what the pattern of connections does. That relational way of thinking is what this course teaches.

Social network analysis treats ties, positions, and structures—not isolated individuals—as the unit of analysis. You will learn to ask and answer questions such as: Which actors occupy central positions? Where do clusters, bridges, and bottlenecks emerge? How do ideas, behaviors, and crises spread? Who holds informal power that formal titles conceal?

This is a hands-on course. Beginning in the first week, you will use R, RStudio, and Gephi in a workflow that recurs throughout the semester: **convert data into a network in R, analyze it, and visualize it in Gephi**. R is our primary environment for reproducible analysis; Gephi is our primary environment for interactive exploration and visual communication. The course uses publicly available networks—including a karate club, a high school, sixteen Florentine families, congressional votes, and a monastery—whose structures can be examined and interpreted directly.

This foundation course emphasizes core concepts, careful data work, and honest interpretation rather

than advanced modeling. It serves as an early course in New College's political methods sequence (**Computational Social Science Track**) and prepares you for more specialized work in social and political networks, computational modeling, and advanced network analysis. More immediately, it gives you two things you can use right away: the ability to read empirical research that uses relational data, and a completed network analysis of your own.

Required Materials

- **Main textbook:** Borgatti, S. P., Everett, M. G., Johnson, J. C., and Agneessens, F. 2022. *Analyzing Social Networks Using R*. SAGE. Cited below as **BEJA**.
- **Visualization supplement:** Cherven, K. 2015. *Mastering Gephi Network Visualization*. Packt. Cited below as **Cherven**. This is a selective reference rather than a cover-to-cover assignment. Because Gephi's interface has changed since 2015, follow the current instructor lab modules when the software differs from the book.
- **Instructor lab modules and R scripts:** posted on Canvas as assigned. Lab modules are required preparation for most applied meetings.
- **Articles:** selected classic or engaging empirical studies, provided on Canvas.
- **Software:** R, RStudio Desktop, and Gephi are free. We will confirm access during the first lab. If the software is not already installed on the lab computers, we will install it together. No programming experience is assumed.

In the schedule, **Main** is required preparation. **Supplement** identifies selective supporting material. **Recommended** readings are optional and most useful when a topic connects to a student's final project.

The Applied Sequence

1. **Units 1–7: data and representation.** Learn the foundations of R; construct node lists, edge lists, and matrices; create network objects; and use Gephi for basic layouts, attribute mapping, and figure export.
2. **Units 8–13: analysis and interpretation.** Compute and interpret network measures in R and develop clear, defensible visualizations in Gephi.
3. **Final project: independent analysis.** Develop the project through a proposal, presentation, and final paper.

Learning Outcomes

By the end of the course, students should be able to:

1. Explain what changes analytically when relations, rather than individual attributes, are the unit of analysis.
2. Distinguish among one-mode, two-mode, directed, undirected, binary, and valued networks.
3. Recognize when a research question calls for network data and construct node, edge, and attribute files in R.
4. Interpret density, components, cliques, communities, centrality, centralization, brokerage, diffusion, prestige, and structural position.
5. Use R to compute and interpret foundational network measures.
6. Use Gephi to produce clear network visualizations using layouts, attributes, partitions, filters, and export controls.
7. Evaluate an empirical network study by connecting its question, node and tie definitions, evidence,

claim, and limitations.

8. Complete a small independent network analysis with a defensible question, transparent workflow, and substantive interpretation.

Course Format

Monday meetings usually develop concepts through explanation, examples, and interpretation. Thursday meetings are usually guided applications in R and Gephi. Because of the Fall 2026 calendar, Units 4, 9, and 13 meet once as focused integrated sessions, and Thursday, October 22 is a College no-class day.

Evaluation

Component	Weight	Due
Participation and discussion	10%	Ongoing
Data-to-Network Assignment	15%	Thu, Sep 24
Midterm examination	20%	Thu, Oct 1
Final project: proposal	10%	Mon, Oct 26
Final project: presentation	15%	Nov 30 / Dec 3
Final project: paper and reproducibility materials	30%	Fri, Dec 11

Participation

Both meetings involve discussion and applied work. The participation grade reflects consistent attendance, preparation, and constructive engagement.

Data-to-Network Assignment (15%, due Thursday, September 24)

Select data on a topic that genuinely interests you—politics, campus life, sports, music, online communities, history, or another approved subject—that contains relationships but is not yet organized as a network. Explain the relational question, define the nodes and ties, construct node and edge lists in R, compute two or three basic measures, and produce one Gephi figure. Submit the documented analysis with a 2–3-page explanation.

Midterm (20%, Thursday, October 1)

The in-class, paper-based examination tests concepts, data understanding, and interpretation rather than live coding. It includes short conceptual questions, interpretation of measures and figures, short excerpts of R code and output, and diagnosis of problems in edge lists and matrices. It covers Units 1–6 in depth and Unit 7's projection material at the conceptual level.

Final Project (55% total)

Complete a small independent network analysis using public, archival, classroom, or approved hand-collected data. The project must include a research question, explicit node and tie definitions, a transparent R workflow, at least two network measures, at least one Gephi figure, and a final paper of no more than 10 pages, including all content.

- **Proposal (10%, due Monday, October 26).** Submit 1–2 pages stating the research question, data source, node and tie definitions, and planned measures. Explain why the proposed network representation fits the question: if the data is not yet in network form, show how you will convert it; if it is already a network, explain how it was obtained and what your analysis will add.

- **Presentation (15%, November 30 and December 3).** Present for approximately eight minutes, followed by two minutes of questions. The format may be adjusted to enrollment.
- **Paper and reproducibility materials (30%, due Friday, December 11).** Submit the paper, R script, cleaned or source data when redistribution is permitted, a README defining nodes and ties, and either the Gephi project or a record of the relevant visualization and export settings.

New College Course Designations and Narrative Evaluation

New College records a course designation and a narrative evaluation addressing conceptual understanding, data management, interpretation, visualization, and engagement. The final percentage converts as follows:

Final percentage	NCF designation
90–100%	Strong Satisfactory (Sat+)
80–89.99%	Satisfactory (Sat)
70–79.99%	Marginal Satisfactory (Sat–)
Below 70%	Unsatisfactory (Unsat)

Essential Course Policies

- **Attendance.** Because the course combines conceptual discussion with live data work, regular attendance matters. More than two unexcused absences may affect the participation grade.
- **Late work.** The assignment and proposal lose one-third of a letter grade per class meeting late unless an extension has been approved in advance.
- **Academic integrity and collaboration.** Follow the [New College Academic Honor Code](#). Discussion and troubleshooting are encouraged, but submitted code and writing must reflect your own work.
- **Use of AI.** AI tools may be used for brainstorming, debugging, and explanation, but you remain responsible for checking code, understanding methods, documenting your workflow, and writing your own interpretations. Submitting unverified AI-generated work as your own analysis is not acceptable.
- **Ethics of network data.** Network data can reveal information about more than one person through a single tie. If a final project uses hand-collected data, including data about classmates, students must obtain informed consent, anonymize identities in submitted and presented work, and receive instructor approval before collecting data.
- **Support and communication.** For accommodations, contact the [Accessible Learning Center](#) and the instructor early in the semester. Statistics and R tutoring is available through the [Academic Resource Center](#). Course materials and updates appear on Canvas; check Canvas and New College email regularly.

Course Schedule

Each unit lists Class 1 and Class 2 when two meetings occur. **Main** is required preparation; **Supplement** is selective support reading; **Recommended** is optional.

Part I — Foundations of Relational Thinking

Unit 1 — Introduction: Networks and Relational Thinking

What changes analytically when we treat relations, rather than isolated individuals, as the primary unit of observation?

Mon, Aug 17

Course orientation: Why networks?

Main: Course syllabus

Recommended: Watts, Duncan J. 2004. "The 'New' Science of Networks." *Annual Review of Sociology* 30: 243–270.

Social network analysis through three public-data cases: the Medici marriage and business networks, high-school contact and friendship networks, and congressional roll-call agreement • relations as the unit of analysis • the R-to-Gephi workflow • course assessments and progression

Thu, Aug 20

Orientation to R, RStudio, and Gephi

Main: Instructor Lab Module 1 (Canvas)

Supplement: BEJA Ch. 5, §§5.1–5.2 (the R program)

R and RStudio: console, scripts, projects, folders, and packages • Gephi: importing a network, applying a layout, and exporting a figure • the relationship between reproducible analysis and visual exploration

Data: *Les Misérables* co-appearance network (Gephi sample dataset)

Unit 2 — The Language of Networks

What are nodes, ties, and graphs—and how does tabular data become network data?

Mon, Aug 24

Graphs, ties, and the goals of network analysis

Main: BEJA Ch. 1, §§1.1–1.7; Ch. 2, §§2.1–2.3

Supplement: Cherven Ch. 1 ("A network graph analysis primer"; "Paths and connectivity")

Graphs, nodes, edges, dyads, and paths • directed and undirected ties • binary and valued ties • node sets, tie definitions, and relational research questions

Thu, Aug 27

From edge list to network object

Main: Instructor Lab Module 2 (Canvas)

Supplement: BEJA Ch. 5, §§5.3–5.5 (data storage, importing, and transformation)

Edge lists and adjacency matrices • creating an *igraph* object • checking identifiers, duplicate ties, self-loops, missing values, and attributes

Data: *Zachary karate club*, built into *igraph* as `make_graph("Zachary")`

Unit 3 — Attributes, Homophily, and Interpretation

How do actors' attributes help us interpret network structure?

Mon, Aug 31

Birds of a feather: attributes and homophily

Main: McPherson, M., L. Smith-Lovin, and J. M. Cook. 2001. "Birds of a Feather: Homophily in Social Networks." *Annual Review of Sociology* 27: 415–444 (read pp. 415–429; skim the rest).

Supplement: BEJA Ch. 3, §3.6 (actor attributes); skim §§3.1–3.5 for orientation

Social, economic, psychological, and geographic attributes • homophily and sorting • structural and compositional interpretations • the difference between observing similarity and explaining it

Thu, Sep 3

Attribute overlays and the first R-to-Gephi round trip

Main: Instructor Lab Module 3 (Canvas)

Supplement: BEJA Ch. 7, §7.3 (embedding node attributes); Cherven Ch. 4 ("Clustering and homophily")

Merging actor attributes into a network object • diagnostic visualization in R • exporting an attributed network to Gephi • appearance mappings and figure export

Data: Zachary karate club with faction attributes, available as `data("karate", package = "igraphdata")`

Part II — Cohesion, Cleavages, and Affiliation Structures

Unit 4 — Cohesion, Density, and Connectedness (single integrated session)

Under what structural conditions do actors function as a cohesive group?

Mon, Sep 7

Labor Day — no class.

Thu, Sep 10

Cohesion, density, and components

Main: BEJA Ch. 10, §§10.1–10.3; review Ch. 2, §2.3 (paths and components)

Supplement: Cherven Ch. 6 ("Graph density"; "Average path length"; "Connected components")

Recommended: Moody, James, and Douglas R. White. 2003. "Structural Cohesion and Embeddedness." *American Sociological Review* 68(1): 103–127.

Density and average degree • reachability, components, isolates, and fragmentation • cohesion as a basis for coordination • why equal density can conceal different component structures • limits of cross-network density comparisons

Data: SocioPatterns high-school contact network (open temporal contacts; aggregated in the instructor script for this unit)

Unit 5 — Cohesive Subgroups and Community Detection

How do we detect dense pockets of actors who "stick together"?

Mon, Sep 14

Cohesive subgroups and community detection

Main: BEJA Ch. 11, §§11.1–11.4 (cliques, Girvan–Newman, and modularity optimization)

Recommended: Moody, James. 2001. "Race, School Integration, and Friendship Segregation in America." *American Journal of Sociology* 107(3).

Cliques and cohesive subgroups • communities and modularity • resolution, stability, and algorithm choice • communities as analytical results rather than automatically meaningful social groups

Thu, Sep 17

Finding communities in R and Gephi

Main: Instructor Lab Module 5 (Canvas)

Supplement: Cherven Ch. 6 ("Modularity"); Ch. 7 (partitioning and clustering)

Maximal cliques • Louvain and walktrap partitions • modularity-based partitioning in Gephi • how layout and color can clarify or exaggerate community structure

Data: SocioPatterns high-school contact network with public class metadata

Unit 6 — Affiliation Networks and Two-Mode Thinking

How do organizations, events, or collective choices connect actors to one another?

Mon, Sep 21

Actors and events: bipartite logic

Main: BEJA Ch. 13, §§13.1 and 13.4 (two-mode data and bipartite networks)

Recommended: Borgatti, Stephen P., and Martin G. Everett. 1997. "Network Analysis of 2-Mode Data." *Social Networks* 19(3).

Two-mode networks and incidence matrices • affiliation as relational infrastructure • individuals and events in the Southern Women study • legislators and roll-call votes as a political affiliation structure

Thu, Sep 24

Assignment due

Building two-mode data from a public CSV

Main: Instructor Lab Module 6 (Canvas)

Supplement: BEJA Ch. 4, §§4.6–4.7 (archival and electronic data sources)

Representing bipartite data in R • constructing and validating an incidence matrix • distinguishing one-mode and two-mode structures

Data: Davis's Southern Women, available as `data("davis", package = "latentnet")`; Voteview member-vote data (public CSV; single-congress extract on Canvas)

Unit 7 — Projection / Midterm

What is gained and lost when shared affiliations are converted into actor-to-actor ties?

Mon, Sep 28

Projection and its interpretive traps

Main: BEJA Ch. 13, §§13.2–13.3 (converting two-mode to one-mode)

Actor and event projections • weighted and binary projections • tie inflation • projecting legislators from shared roll-call positions • why co-affiliation should not automatically be interpreted as collaboration

Thu, Oct 1

Midterm

Midterm examination

In class, on paper; no live coding. Covers Units 1–6 in depth and Unit 7's projection material at the conceptual level. A review guide will be posted on Canvas.

Oct 5–9

Fall Break — no class.

Mon, Oct 12

Columbus Day/Indigenous People's Day — no class.

Special Topic (no Monday meeting) — Measuring a Network

How much of a network's apparent structure comes from the way ties are observed and recorded?

Thu, Oct 15

Measuring a network

Main: Mastrandrea, Rossana, Julie Fournet, and Alain Barrat. 2015. "Contact Patterns in a High School: A Comparison between Data Collected Using Wearable Sensors, Contact Diaries and Friendship Surveys." *PLOS ONE* 10(9): e0136497.

Supplement: BEJA Ch. 4, §§4.1–4.5 (network data collection)

Sensors, diaries, and friendship nominations as different measures of social relations • missingness, recall, and reporting asymmetry • temporal aggregation • how measurement choices alter density, components, centrality, and substantive conclusions

Data: SocioPatterns high-school contact, diary, friendship, and metadata files (direct public downloads)

Part III — Centrality, Brokerage, and Power

Unit 8 — Centrality and Centralization

How do we distinguish influential actors from networks organized around a center?

Mon, Oct 19

Four ways to be central

Main: BEJA Ch. 9, §§9.1–9.3 (undirected, non-valued centrality); Ch. 10, §10.4 (centralization and core-peripheriness)

Recommended: Freeman, Linton C. 1978–79. "Centrality in Social Networks: Conceptual Clarification." *Social Networks* 1(3).

Degree as activity • closeness as efficiency of reach • betweenness as potential control of flow • eigenvector centrality as connection to the well connected • node centrality and graph centralization

Thu, Oct 22

Mon, Oct 26

Proposal due

Mini-Classes for Spring 2027 — no class.

Computing and comparing centralities

Main: Instructor Lab Module 8 (Canvas)

Supplement: Cherven Ch. 6 (centrality measures and interpretation); Ch. 7 (numeric ranking, read alongside the current Appearance workflow in the instructor module)

Computing four centrality measures in R • comparing rankings and interpreting disagreement across measures • graph centralization • Gephi Appearance controls for representing centrality without confusing size with importance

Data: Florentine family marriage and business networks, loaded through `data("florentine", package = "ergm")`

Unit 9 — Bridges, Bottlenecks, and Vulnerability (single integrated session)

Where are a network's structural weak points, and what happens when they fail?

Thu, Oct 29

Finding and testing structural weak points

Main: Instructor Module 9: Concept and Lab (Canvas); Albert, Réka, Hawoong Jeong, and Albert-László Barabási. 2000. "Error and Attack Tolerance of Complex Networks." *Nature* 406: 378–382.

Supplement: Review BEJA Ch. 2, §2.3 (paths and connectivity); Cherven Ch. 6 (connected components)

Bridges and articulation points • component count and largest-component size • random failure and targeted removal • structural vulnerability and assumptions about what flows through a network

Data: Krackhardt kite network, available as `data("kite", package = "igraphdata")`

Unit 10 — Structural Holes and Brokerage

When do actors gain strategic advantage by connecting otherwise separated groups?

Mon, Nov 2

The strength of standing between

Main: BEJA Ch. 8, §§8.1 and 8.6 (ego-network structure, effective size, and constraint); Burt, Ronald S. 2004. "Structural Holes and Good Ideas." *American Journal of Sociology* 110(2): 349–399 (read pp. 349–365 and 388–391).

Supplement: Review BEJA Ch. 9, §9.3 (betweenness)

Brokerage and structural holes • effective size and Burt's constraint • *tertius gaudens* and *tertius iungens* • individual advantage and system-level vulnerability

Thu, Nov 5

Measuring brokerage

Main: Instructor Lab Module 10 (Canvas)

Supplement: Cherven Ch. 6 ("Neighborhood overlap and embeddedness"); Ch. 5 (topology filters)

Computing constraint and effective size • distinguishing centrality from brokerage • substantive interpretation of low-constraint positions • layouts and filters that reveal bridging without overstating it

Data: Florentine family networks, loaded through `data("florentine", package = "ergm")`

Part IV — Diffusion, Prestige, and Hierarchy

Unit 11 — Diffusion, Contagion, and Network Process

How does network structure shape who adopts, when they adopt, and how far adoption spreads?

Mon, Nov 9

Simple and complex contagion

Main: Centola, Damon. 2010. "The Spread of Behavior in an Online Social Network Experiment." *Science* 329(5996): 1194–1197.

Supplement: Review BEJA Ch. 1, §1.5 (network variables as explanations)

Recommended: Centola, Damon, and Michael Macy. 2007. "Complex Contagions and the Weakness of Long Ties." *American Journal of Sociology* 113(3): 702–734.

Diffusion as a network process • simple and complex contagion • exposure, reinforcement, thresholds, and temporal sequence • when long ties accelerate or inhibit spread

Thu, Nov 12

Watching things spread

Main: Instructor Lab Module 11 (Canvas; includes a pre-written skeleton script)

Supplement: Cherven Ch. 4 (“Contagion and diffusion”); Ch. 8 (“Dynamic Networks”)

Discrete-time diffusion • SI and threshold processes • cascades on random, small-world, and clustered networks • dynamic views and timelines in Gephi

Data: Networks generated reproducibly with `igraph` random-graph and small-world functions using the instructor script

Unit 12 — Directed Networks and Structural Prestige

How do asymmetric ties reveal deference, authority, and status?

Mon, Nov 16

When ties have direction

Main: BEJA Ch. 9, §9.4 (directed, non-valued networks); Ch. 10, §10.2.2 (reciprocity)

Recommended: Fowler, James H., et al. 2007. “Network Analysis and the Law.” *Political Analysis* 15(3): 324–346.

Directed ties • reciprocity and asymmetry • indegree and outdegree • structural prestige and its relationship to formal status

Thu, Nov 19

Prestige in practice

Main: Instructor Lab Module 12 (Canvas)

Supplement: Cherven Ch. 6 (“In-degree centrality”; “Out-degree centrality”)

Directed adjacency matrices • computing indegree, outdegree, and reciprocity • comparing network-derived prestige with external indicators of rank

Data: **Sampson monastery liking network**, third observation `samplk3`, loaded through `data("samplk", package = "ergm")`

Unit 13 — Positions and Informal Hierarchy (single integrated session)

Can network analysis recover latent positions and informal hierarchies that formal charts conceal?

Mon, Nov 23

Positions and informal hierarchy

Main: BEJA Ch. 12, §§12.1–12.2 (introduction and structural equivalence); Padgett, John F., and Christopher K. Ansell. 1993. “Robust Action and the Rise of the Medici, 1400–1434.” *American Journal of Sociology* 98(6): 1259–1319 (read pp. 1259–1277; skim the remaining case analysis).

Supplement: Cherven Ch. 3 (radial, concentric, and layered layouts)

Recommended: BEJA Ch. 12, §12.3 (profile similarity)

Formal and informal hierarchy • how centrality, community, and structural position answer different questions • structural equivalence at the intuitive level • the Medici network as a theory of robust political action • why a hierarchical-looking layout does not establish hierarchy

Data: **Florentine family marriage and business networks**, loaded through `data("florentine", package = "ergm")`

Thu, Nov 26–27

Thanksgiving Break — no class.

Unit 14 — Capstone Presentations

Moving from metric calculation to defensible social-science argument.

Mon, Nov 30

Presentations

Presentations, first half

Approximately half of the class presents; presentation order and timing will be adjusted to enrollment.

Thu, Dec 3

Presentations

Presentations, second half

Remaining presentations • synthesis across cohesion, affiliation, centrality, brokerage, diffusion, prestige, and hierarchy

Final Evaluation Period

Mon–Tue, Dec 7–8

Reading/Hurricane Make-up Days — no regular class unless the College announces a required make-up arrangement.

Wed–Fri, Dec 9–11

*Final evaluation period. **Final paper due Friday, December 11.***

Paper due Dec 11